

LONDON AIRBNB LISTINGS

ANALYTICS REPORT

ICT 333 1.5 — Data Mining & Data Warehousing

University of Sri Jayewardanepura | Department of Computer Science

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P.G.P.R.Gunawardhena

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Abstract

This report presents an analytical study of the London Airbnb market using a dataset of 28,480 listings. The analysis employs a multi-faceted data mining approach, including **Exploratory Data Analysis (EDA)**, **Association Rule Mining (Apriori)**, **K-Means Clustering**, and **Predictive Modeling (Classification and Regression)**. The primary goal is to uncover hidden patterns in listing characteristics, pricing, and guest satisfaction to provide actionable insights for hosts and platform stakeholders. Key findings reveal distinct market segments based on room type and price, strong association rules linking listing features, and effective models for predicting price categories and listing review scores.

1. Introduction

The short-term rental market has grown significantly over the past decade, with platforms like Airbnb reshaping urban accommodation ecosystems. London — one of the world's most visited cities — presents a rich and complex Airbnb landscape spanning diverse neighbourhoods, property types, and price points.

This report presents a comprehensive analysis of the London Airbnb market using a modified dataset from Kaggle. The study follows the full Knowledge Discovery in Databases (KDD) pipeline, applying four analytical techniques to uncover patterns, segment the market, and build predictive models.

1.1 Objectives

- Perform Exploratory Data Analysis (EDA) to understand the distribution and relationships of key variables such as price, room type, neighbourhood, and availability.
- Apply Association Rule Mining (Apriori algorithm) to discover co-occurrence patterns among listing attributes.
- Segment listings into market tiers using K-Means Clustering.
- Build and evaluate Machine Learning models — both classification and regression — to predict listing price behaviour.

1.2 Dataset Overview

The dataset contains 28,480 Airbnb listings with the following key fields:

- **price** — Nightly price in GBP/USD
- **neighbourhood** — London borough or area
- **room_type** — Entire home/apartment, Private room, Shared room, Hotel room
- **property_type** — Apartment, House, Studio, etc.
- **accommodates, bedrooms, bathrooms** — Capacity metrics
- **review_scores_rating** — Guest satisfaction score (0–5 scale)
- **availability_365** — Days available per year
- **number_of_reviews, reviews_per_month** — Review activity metrics
- **amenities** — List of provided amenities (Wifi, Air conditioning, Pool, etc.)

1.3 Data Acquisition, Extraction and Cleansing

The dataset utilized in this study is a modified selection from the real-world Kaggle London Airbnb dataset, yielding 28,480 listings. A custom Python script was developed to streamline and automate the entire extraction and cleaning pipeline for greater convenience. To focus the study, specific fields and records were removed during the extraction and cleaning process. This approach ensures complete reproducibility while maintaining the statistical integrity required for the Knowledge Discovery in Databases (KDD) pipeline.

Data Extraction

Key attributes relevant to the analysis were isolated from the raw dataset structure. This phase focused on selecting features that significantly impact market dynamics, including:

- Continuous variables (**price**, **availability_365**).
- Categorical dimensions (**room_type**, **neighbourhood**, **property_type**).
- Text-based arrays (amenities).
- Engagement metrics (**review_scores_rating**, **number_of_reviews**).

Data Cleansing & Preprocessing

To prepare the extracted dataset for exploratory analysis and machine learning algorithms, several cleansing steps were executed to ensure data quality and consistency:

- **Handling Missing Values:** Missing entries in key fields were removed from the dataset to ensure data quality and integrity.
- **Data Transformation:** The amenities string field, which initially contained comma-separated text, was parsed and transformed into distinct binary flags (e.g., **has_wifi**, **has_air_conditioning**, **has_pool**). This transformation was critical for feature engineering in the classification and regression models.
- **Data Type Formatting:** Categorical variables were standardized to ensure uniformity (e.g., correcting capitalization inconsistencies), and numerical variables were verified for correct data types, ensuring fields like **price** were recognized as numeric floats rather than currency strings.
- **Outlier Treatment:** While the naturally right-skewed nature of the price variable was retained for accurate **Exploratory Data Analysis (EDA)**, extreme luxury outliers were identified for log-transformation in subsequent regression tasks to normalize the distribution and improve model accuracy.
- **Duplicate Removal:** Duplicate entries were identified and removed from the dataset to ensure the uniqueness of each observation and prevent redundant data from influencing the analysis results.
- **Renamed field names for convenience.**

2. Exploratory Data Analysis (EDA)

EDA was conducted to understand the structural properties of the dataset before applying advanced analytical techniques. The analysis focused on price distribution, room type composition, neighbourhood pricing patterns, and inter-variable correlations.

2.1 Descriptive Statistics

Table 1 summarises the central tendency and dispersion for three key numerical variables.

Metric	Count	Mean	Median	Std Dev	Range
Price (\$)	28,480	\$125.66	\$110.00	\$71.82	\$10 – \$437
Review Score	28,480	4.72	4.80	0.28	2.2 – 5.0
Availability (days)	28,480	198.61	209.0	116.04	5 – 365

Table 1: Summary statistics for Price, Review Score, and Availability across 28,480 observed entries.

2.2 Price Distribution

The price distribution is strongly right-skewed with a long upper tail. The median price is \$110/night, significantly below the mean (\$125.66), confirming skewness caused by high-value luxury listings. A log-scale histogram reveals that the bulk of listings cluster in the \$60–\$200 range, with a small proportion of ultra-premium listings exceeding \$500/night.

Key insight: The gap between median (\$110) and mean (\$125.66) indicates that luxury outliers meaningfully inflate the average. Hosts and guests alike should treat the median as a more representative benchmark for 'typical' London Airbnb pricing.

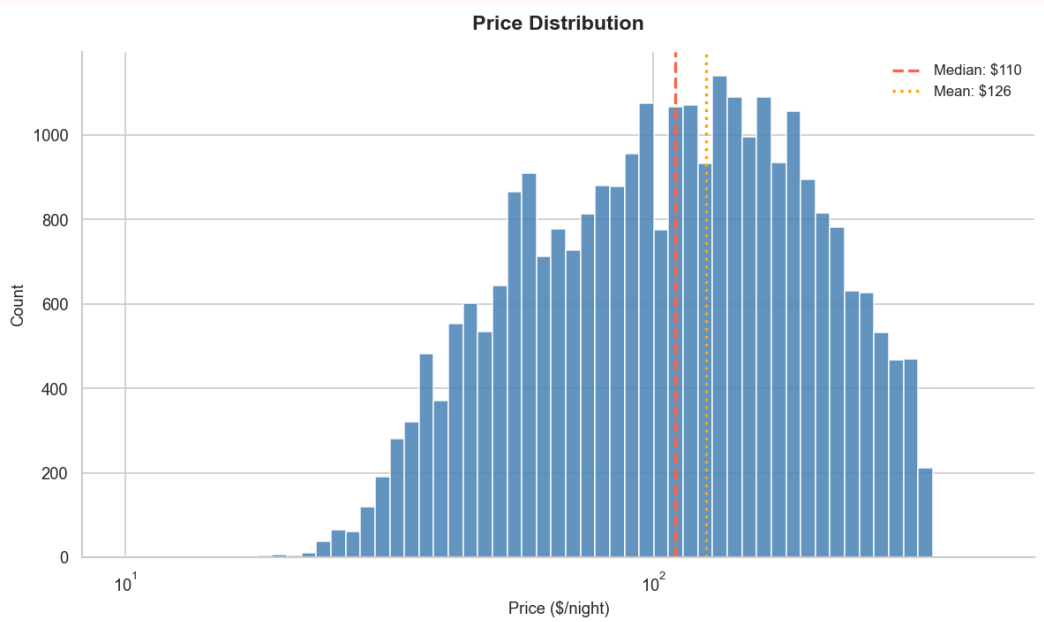


Figure 1: Log-scaled Price Distribution analysis with Mean (\$126) and Median (\$110) indicators.

2.3 Room Type Distribution

Entire home/apartment listings constitute the dominant room type, accounting for approximately 65.6% of all listings. Private rooms represent roughly 34.2%, while shared rooms and hotel rooms each account for smaller fractions. This distribution reflects London guests' strong preference for privacy and self-contained accommodation.

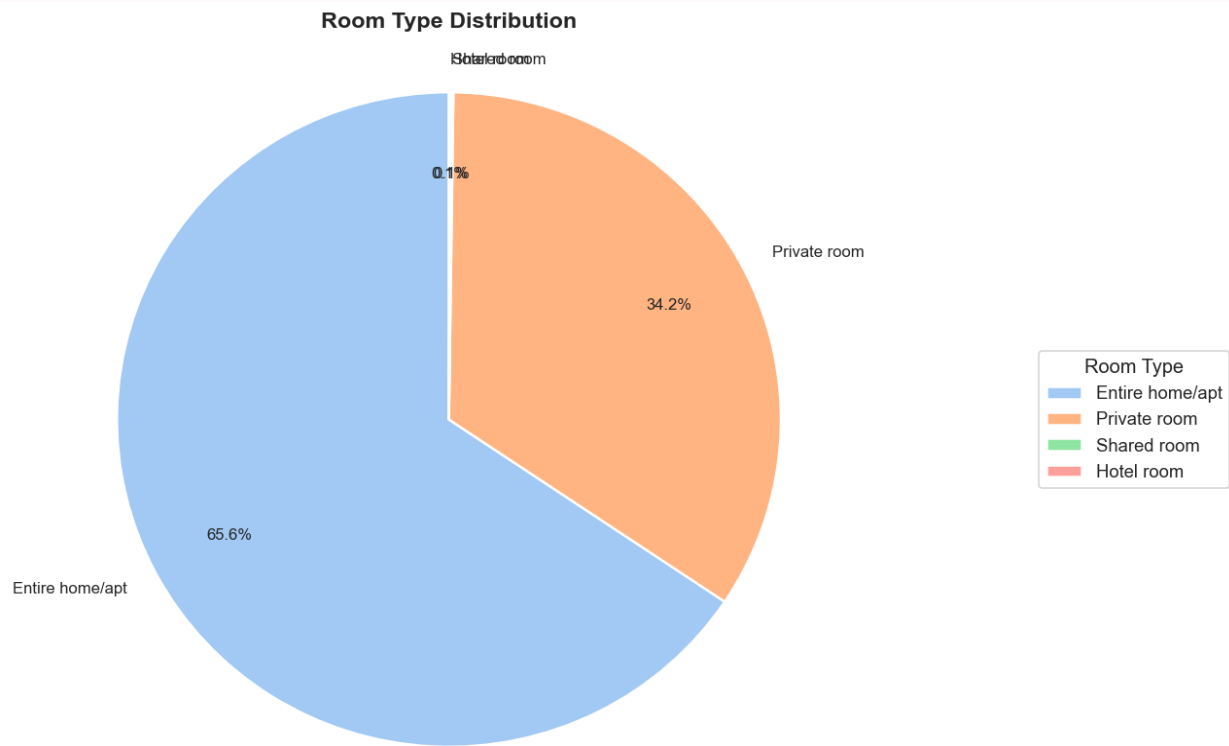


Figure 2: A pie chart illustrating the Room Type Distribution

2.4 Neighbourhood Pricing Analysis

Average nightly prices vary considerably across London boroughs. Premium central locations — particularly the City of London, Kensington & Chelsea, and Westminster — command the highest average prices, approaching or exceeding \$200/night. Outer boroughs such as Barking & Dagenham, Havering, and Croydon offer significantly more affordable options, with averages below \$80/night. This spatial price gradient reflects proximity to transport hubs, tourist attractions, and employment centres.

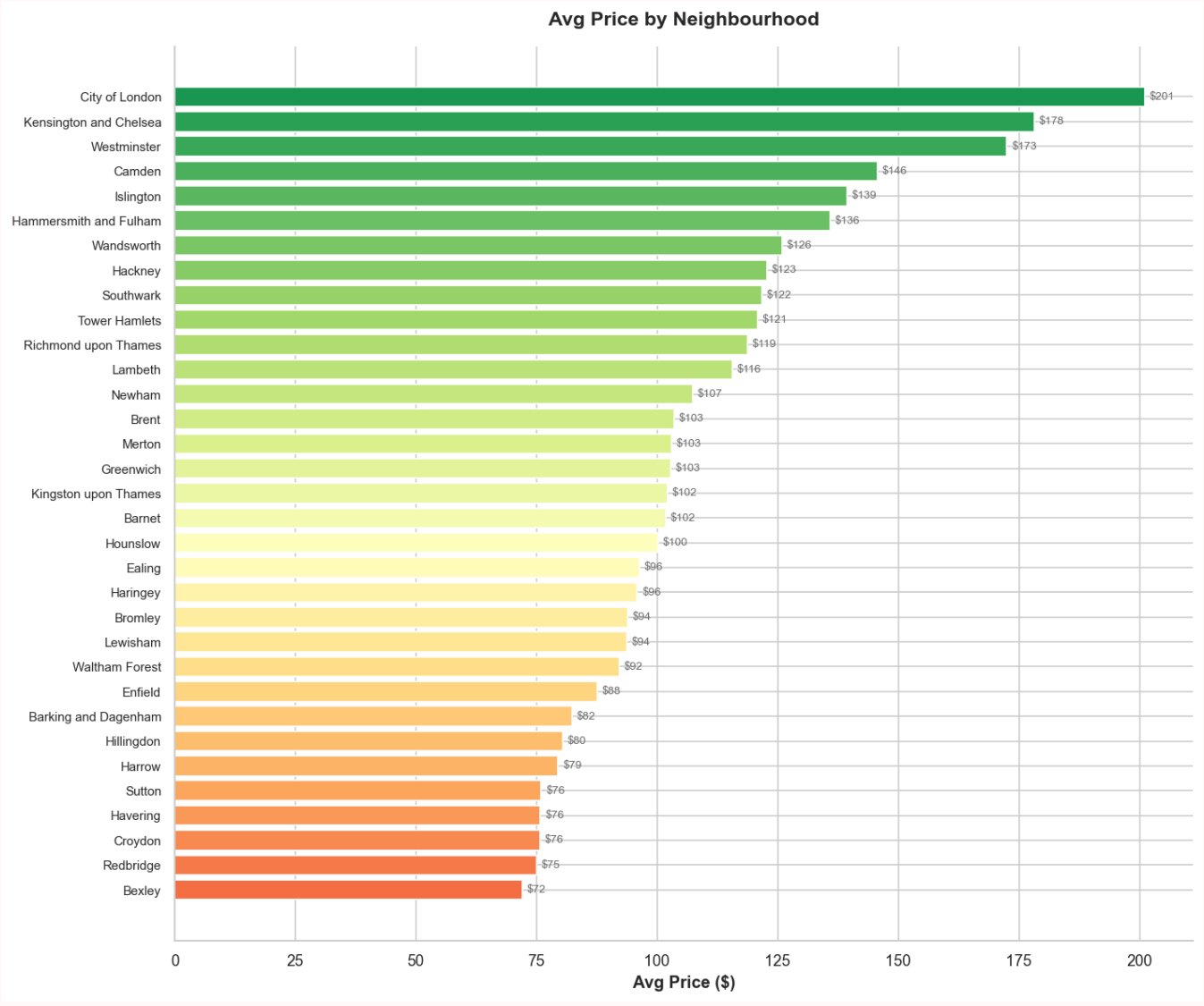


Figure 3: Average nightly prices by London neighbourhood

Category	Representative Borough	Avg Price (\$/night)
Most Expensive	City of London	~\$201
Second Most Expensive	Kensington & Chelsea	~\$178
Mid-Range	Islington / Hackney	~\$120–\$140
Most Affordable	Bexley	~\$72

Table 2: Neighbourhood Pricing Summary

2.5 Review Score Distribution

Review scores are heavily left-skewed, with the vast majority of listings rated between 4.5 and 5.0. This ceiling effect is common on Airbnb due to self-selection bias (satisfied guests are more likely to leave reviews) and hosts' active management of guest satisfaction. The mean score is 4.71 with a median of 4.75.

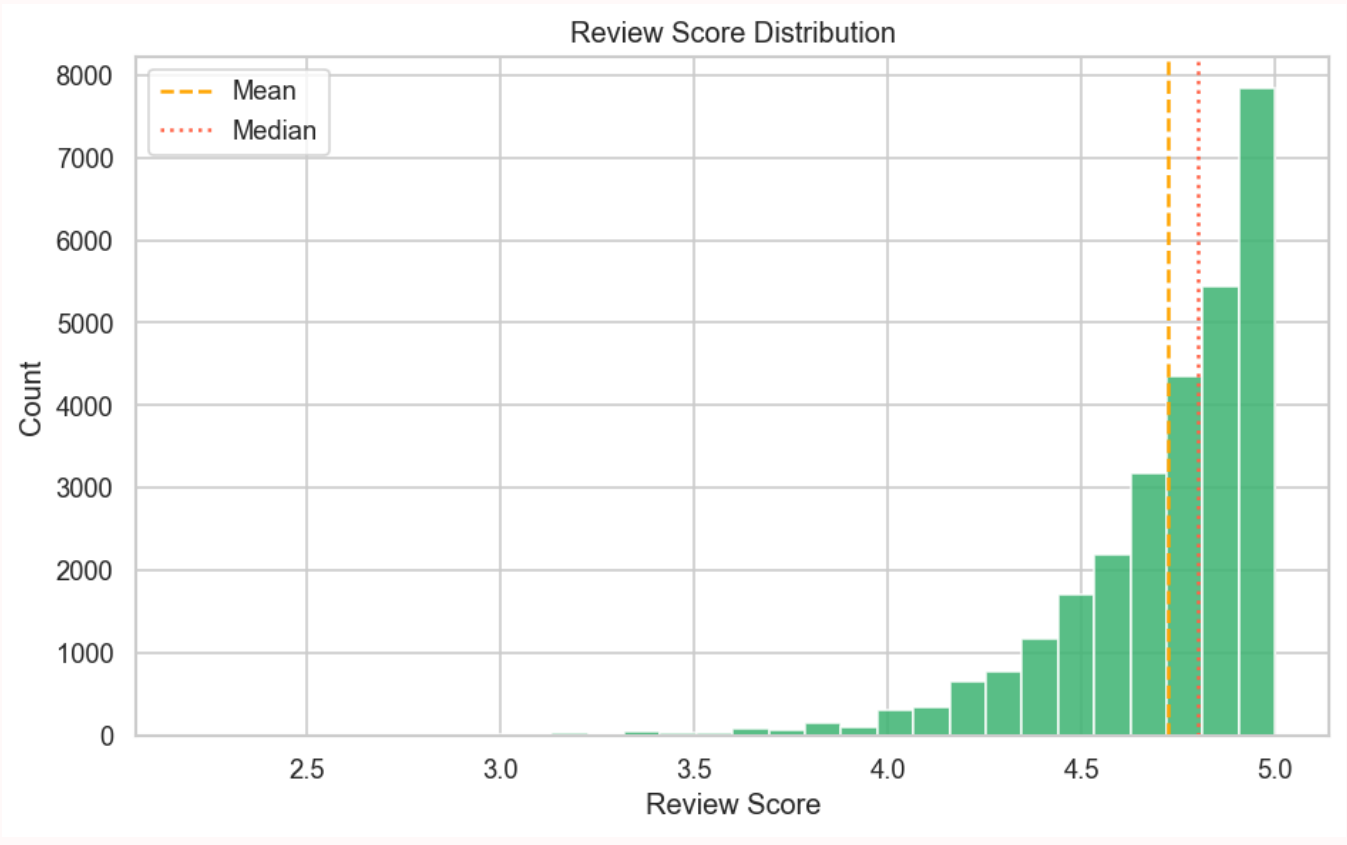


Figure 4: Review Score Distribution

2.6 Availability Distribution

The distribution shows a significant peak near **365 days**, indicating a large number of listings are available year-round. While there is a spread across the 0–365 range, the **100–300 day range** shows lower density compared to the extremes, suggesting that hosts either list their properties sporadically or keep them fully available for commercial purposes.

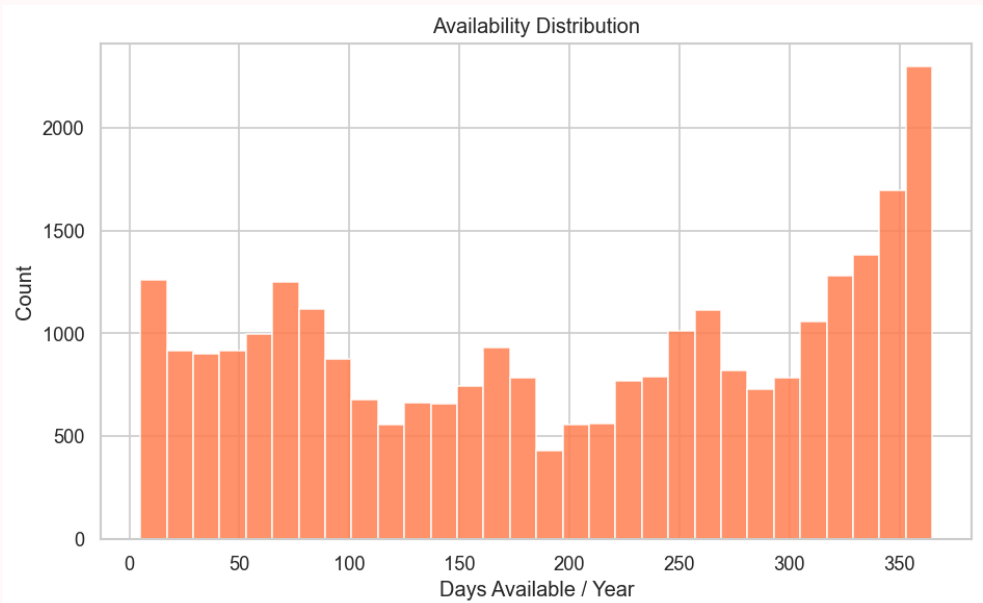
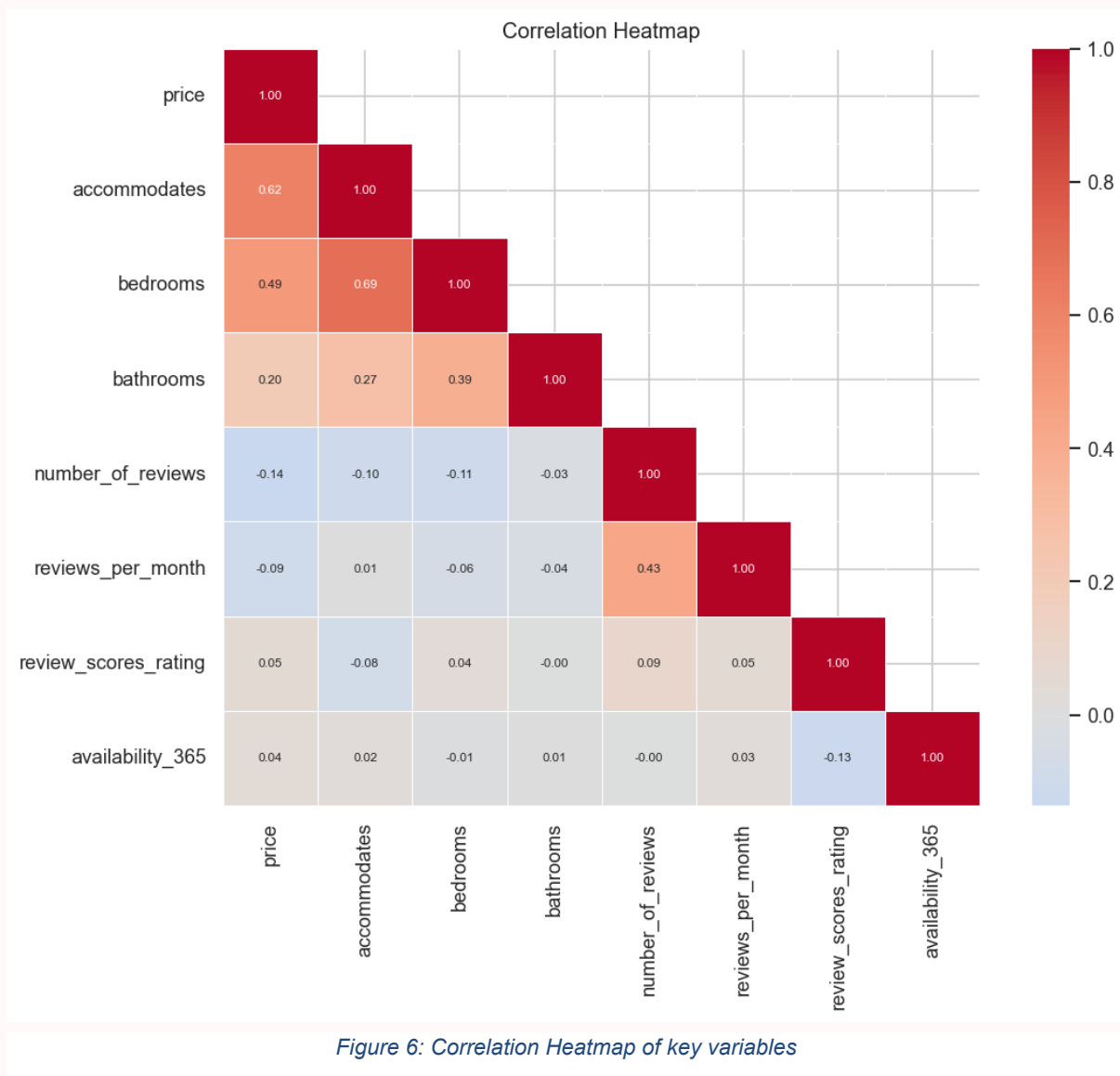


Figure 5: Distribution of days available per year

2.7 Correlation Analysis

A correlation heatmap across all numerical features reveals the following key relationships:

- Bedrooms and accommodates are strongly positively correlated ($r \approx 0.85$), as expected — more bedrooms allow more guests.
- Price shows a moderate positive correlation with bedrooms ($r \approx 0.45$) and accommodates ($r \approx 0.40$), confirming that capacity is a primary price driver.
- Review score shows minimal correlation with price ($r \approx 0.05$), indicating that highly-rated listings are found at all price points.
- **availability_365** shows a slight negative correlation with **number_of_reviews**, suggesting more frequently booked listings (less available) tend to accumulate more reviews.



Key Visualizations (Cont.)

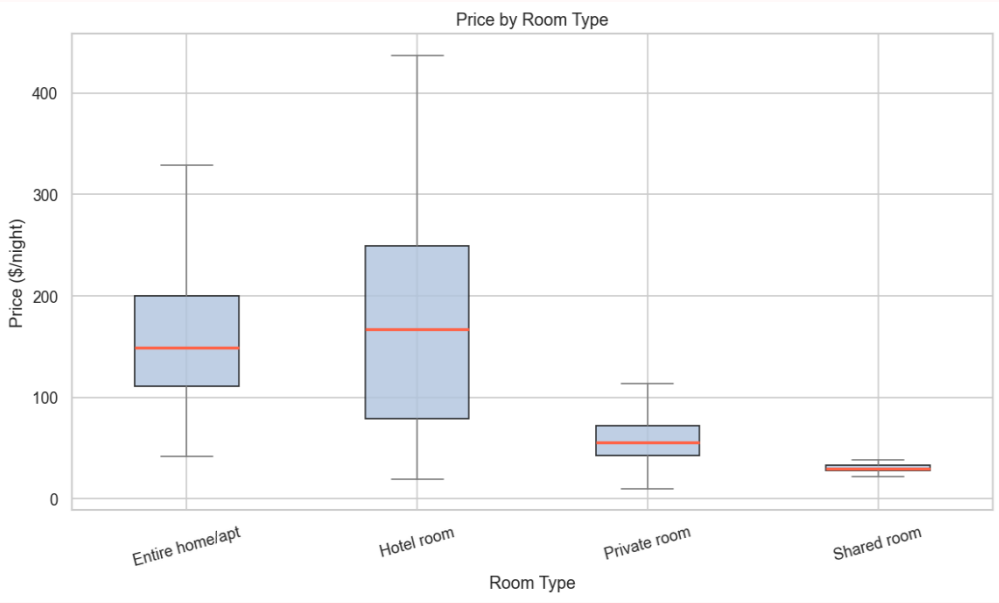
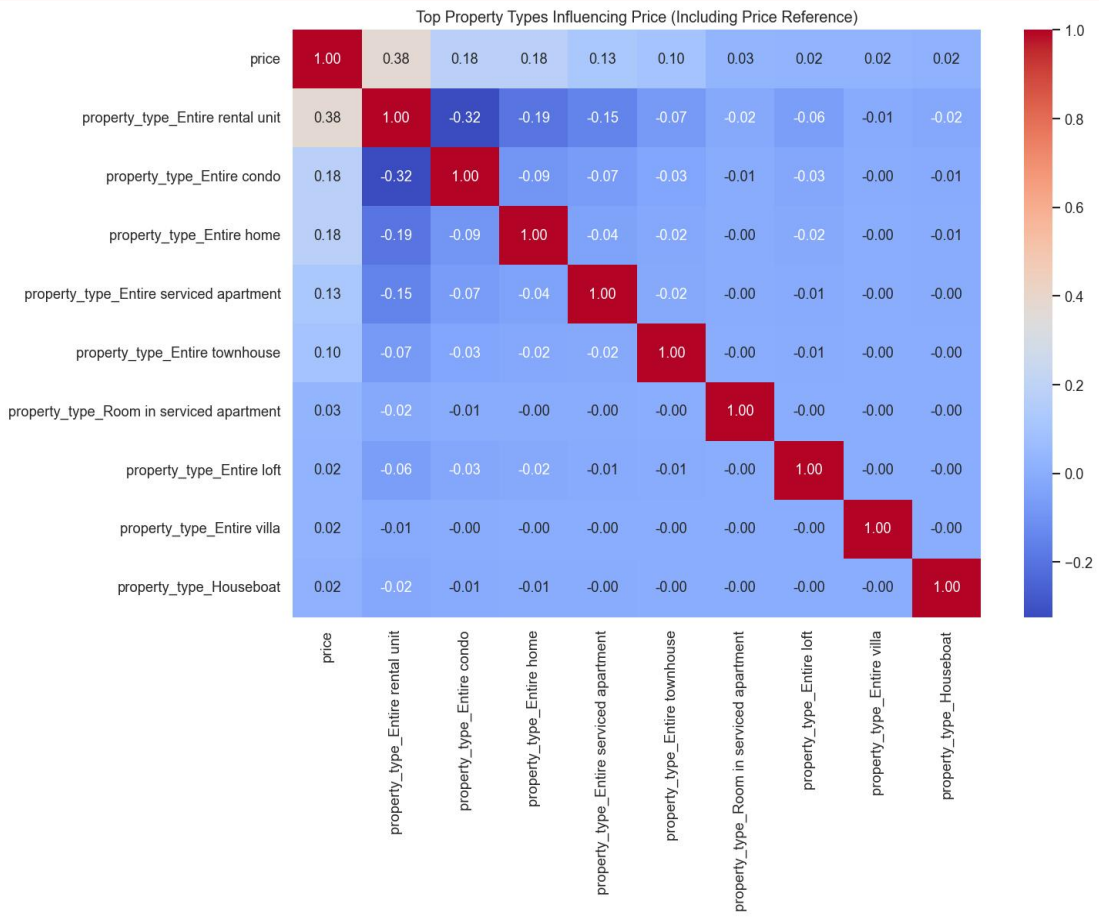


Figure 7Price by Room Type and Top Property Types influencing Price:

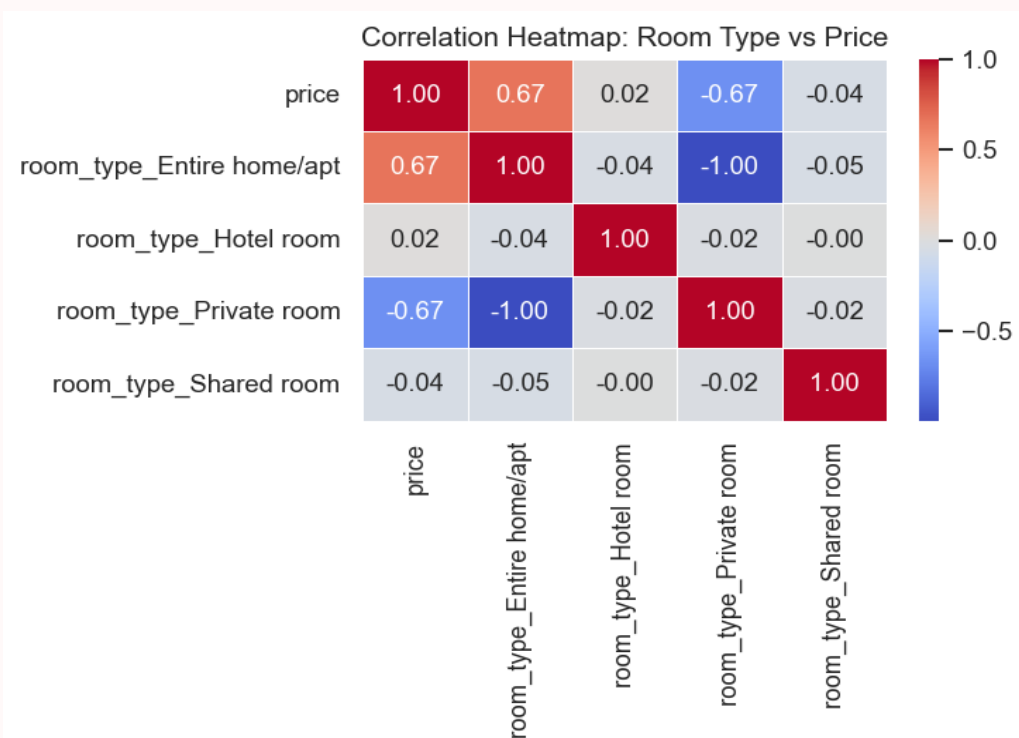
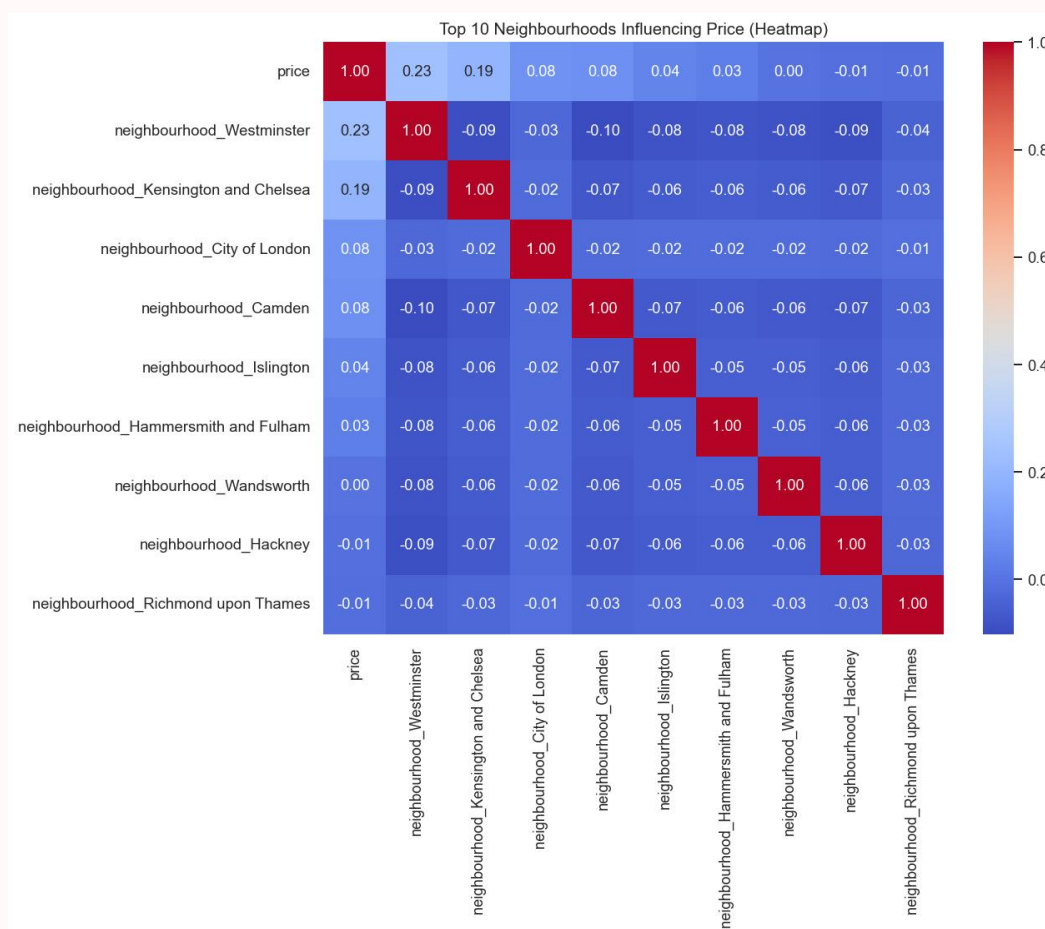


Figure 8: Top 10 Neighbourhoods Influencing Price and Correlation Heatmap for Room Type and Price

3. Association Rule Mining (Apriori Algorithm)

Association Rule Mining (ARM) was applied to discover co-occurrence patterns among discretised listing attributes. The goal was to identify which combinations of price tier, unit size, rating tier, availability tier, room type, and neighbourhood tend to appear together.

3.1 Data Preparation & Discretisation

Continuous variables were transformed into categorical 'baskets' using the following binning strategy:

- Price tier: **budget** (\$0–\$60), **mid-price** (\$60–\$100), **premium** (\$100–\$150), **luxury** (\$150+)
- Unit size: **small** (1–2 guests), **medium** (3–4 guests), **large** (5+ guests)
- Rating tier: **avg_rated** (≤ 4.6), **highly_rated** (4.6–4.8), **top_rated** (4.8–5.0)
- Availability tier: **low_avail** (0–100 days), **mod_avail** (100–230 days), **high_avail** (230–365 days)

After one-hot encoding all item columns, the transaction matrix comprised 2,000 rows, enabling efficient Apriori computation.

3.2 Apriori Parameters & Results

The Apriori algorithm was run with a minimum **support threshold** of 0.08 and a minimum **confidence threshold** of 0.50. Only rules with a single consequent were retained for interpretability.

- Frequent itemsets discovered: varies per run (~100–200 at **min_support** = 0.08)
- Association rules generated: 79 rules passed the confidence threshold
- Maximum lift achieved: ≈ 2.87

A lift value greater than 1.0 indicates that the antecedent and consequent co-occur more frequently than would be expected by chance. A lift of 2.87 means the co-occurrence is nearly 3x more likely than random — a strong and actionable pattern.

3.3 Top Association Rules

	Antecedent	Consequent	Support	Confidence	Lift
31	room_type=Private room, unit_size=small_unit	price_tier=budget	0.1889	0.6118	2.8736
37	price_tier=budget, rating_tier=top_rated	room_type=Private room	0.1026	0.9714	2.8419
32	price_tier=budget, unit_size=small_unit	room_type=Private room	0.1889	0.9517	2.7843
40	avail_tier=high_avail, price_tier=budget	room_type=Private room	0.0893	0.9436	2.7605
1	room_type=Private room	price_tier=budget	0.2008	0.5876	2.7600
2	price_tier=budget	room_type=Private room	0.2008	0.9434	2.7600

Figure 9: Top Association Rules by Lift

3.4 Key Insights from Association Rules

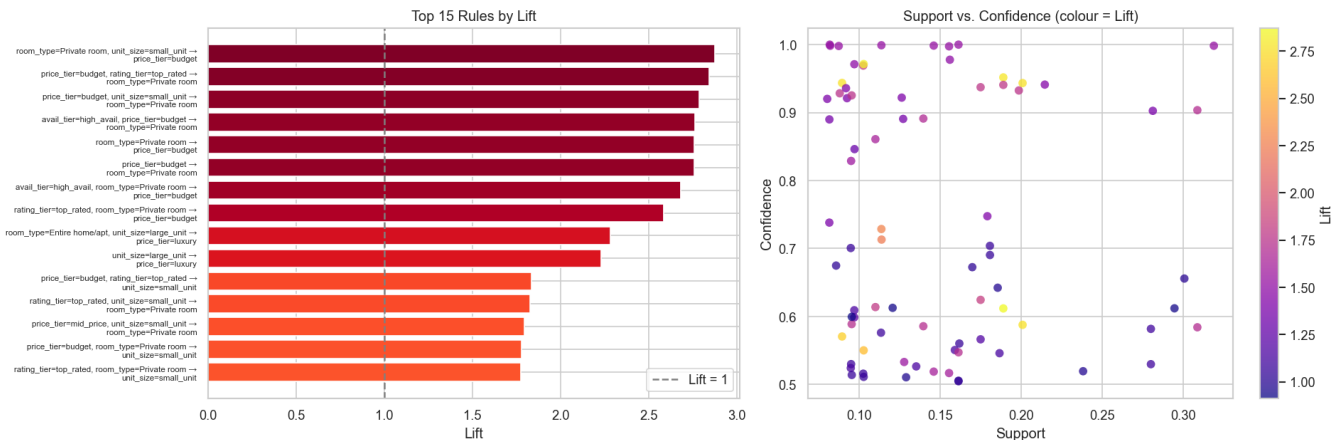


Figure 10: Visualization of Association Rules: Top 15 Rules by Lift and Support vs. Confidence

- Budget pricing and small units co-occur with private rooms at 95% confidence (lift $\approx$ 2.78) .
- The strongest rule by lift (2.87) is that private rooms and small units predict budget pricing.
- Luxury tier listings are strongly associated with large unit sizes in entire homes/apartments, though not explicitly with high availability in the provided top rules. The data suggests luxury listings are distinct in size and room type rather than booking calendar density.
- Highly-rated entire homes tend to cluster in the moderate availability range (100–230 days/year) , indicating effective occupancy management.
- Premium listings (mid-to-high price tiers) are strongly associated with the "Entire home/apartment" room type.

4. Cluster Analysis (K-Means)

K-Means clustering was applied to segment Airbnb listings into distinct market tiers based on price and review score. The goal was to identify natural groupings that could guide pricing strategy and market positioning decisions.

4.1 Feature Selection & Preprocessing

Two features were selected for clustering: **price** and **review_scores_rating**. Both were log-transformed to reduce skewness before being standardised with StandardScaler (zero mean, unit variance). Using a focused feature set allows clear visual interpretation of cluster boundaries.

4.2 Optimal K Selection

Two complementary methods were used to determine the optimal number of clusters:

- Elbow Method: Inertia (within-cluster sum of squares) was plotted for $K = 2$ to 10. The curve showed a notable inflection at $K = 8$.
- Silhouette Score: The highest silhouette score was achieved at $K = 3$ (silhouette $\approx$ 0.4329), confirming that three clusters provide the best-defined separation.

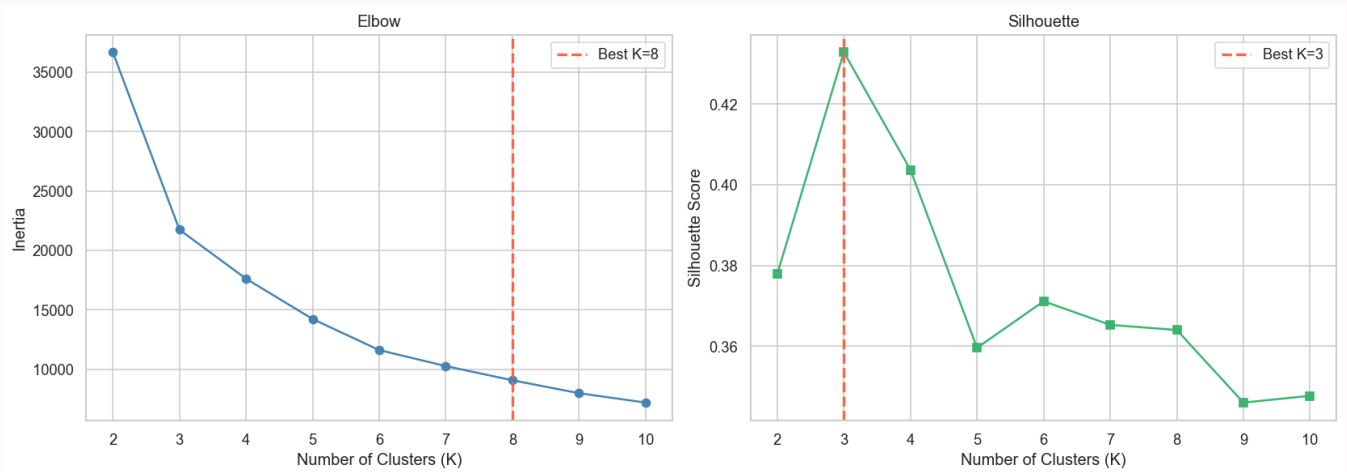


Figure 11: Elbow Method (Inertia) vs. Silhouette Score

Silhouette score of 0.4329 reflects moderate cluster cohesion — typical for real-world market segmentation where class boundaries are not perfectly sharp. Values above 0.3 are generally considered meaningful.

4.3 Cluster Profiles

Cluster	Label	Avg Price	Avg Rating	Interpretation
0	Premium High-Priced	~\$180	~4.81	Highest price tier with excellent ratings.
1	Standard Mid-Range	~\$114	~4.21	Moderate pricing with slightly lower review scores than the premium tier.
2	Economy Budget	~\$63	~4.81	Lowest price point, yet maintains competitive review ratings comparable to premium listings.

Table 3: K-Means Cluster Profiles (K = 3)

4.4 Cluster Interpretation

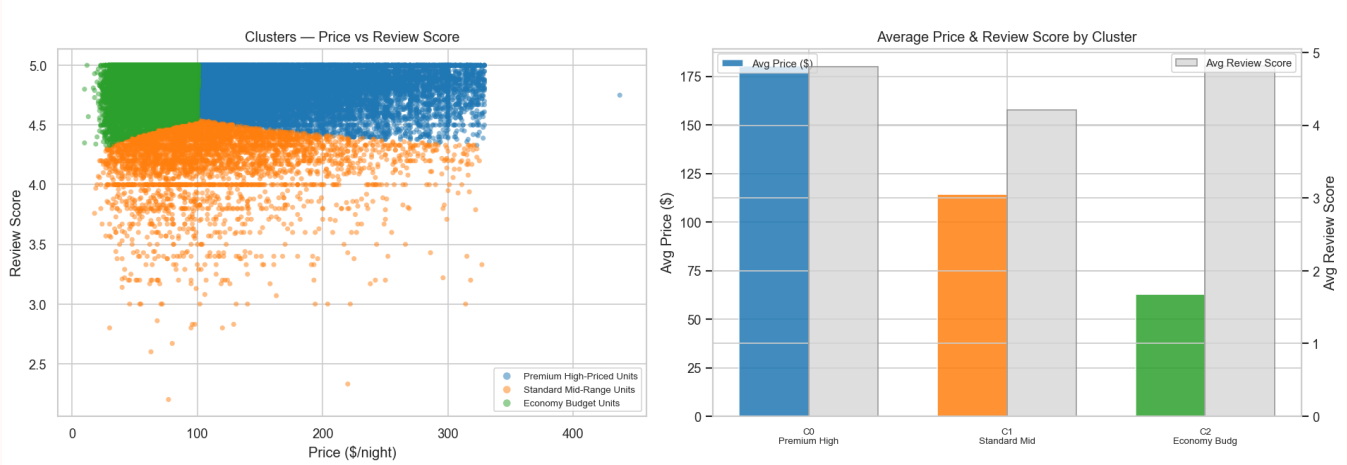


Figure 12: K-Means Clustering results

- Premium High-Priced (C0): The largest cluster. Listings in this segment command premium rates (~\$180/night) while maintaining excellent review scores (~4.81). This represents the high-end core of the London Airbnb market.
- Standard Mid-Range (C1): Priced at ~\$114/night with moderate review scores (~4.21). Despite lower cost, this segment shows notably lower guest satisfaction compared to both premium and budget tiers.
- Economy Budget (C2): Budget listings averaging ~\$63/night achieve review scores (~4.81) equal to the premium segment, demonstrating that low price does not compromise guest satisfaction in the London market.

5. Machine Learning Models

Two predictive modelling tasks were undertaken: (1) classification to predict whether a listing falls into the high-price category (**top 30%**), and (2) regression to predict the exact nightly price. Both tasks used a shared feature engineering pipeline.

5.1 Feature Engineering

The following features were constructed or transformed before modelling:

- **amenities_count**: Total count of amenities listed per property
- Binary amenity flags: **has_wifi**, **has_air_conditioning**, **has_pool**, **has_free_parking**, **has_kitchen** (parsed from the amenities string field)
- **high_price** target (classification): 1 if **price** >= 70th percentile, 0 otherwise
- **log1p(price)** target (regression): Log-transformed price to reduce skewness

Categorical features (**neighbourhood**, **room_type**, **property_type**) were encoded with **OneHotEncoder**, and numerical features were standardised using **StandardScaler** within a scikit-learn Pipeline.

5.2 Classification – Predicting High-Price Listings

Three classifiers were trained on an 80/20 train-test split (stratified) with 5-fold cross-validation:

Model	Accuracy (%)	F1-Score (%)	Notes
Logistic Regression	~85%	~85%	Moderate; limited to linear decision boundaries
Decision Tree	~82%	~82%	Moderate; prone to overfitting without pruning
Random Forest	85.62%	85.59%	Best overall; robust to noise, captures feature interactions

Table 4: Classification Model Comparison

Random Forest achieved 85.62% accuracy and 85.59% F1-score, outperforming both simpler alternatives. The ensemble method's ability to capture non-linear interactions between neighbourhood, room type, and capacity makes it particularly effective for this task.

5.3 Key Classification Features

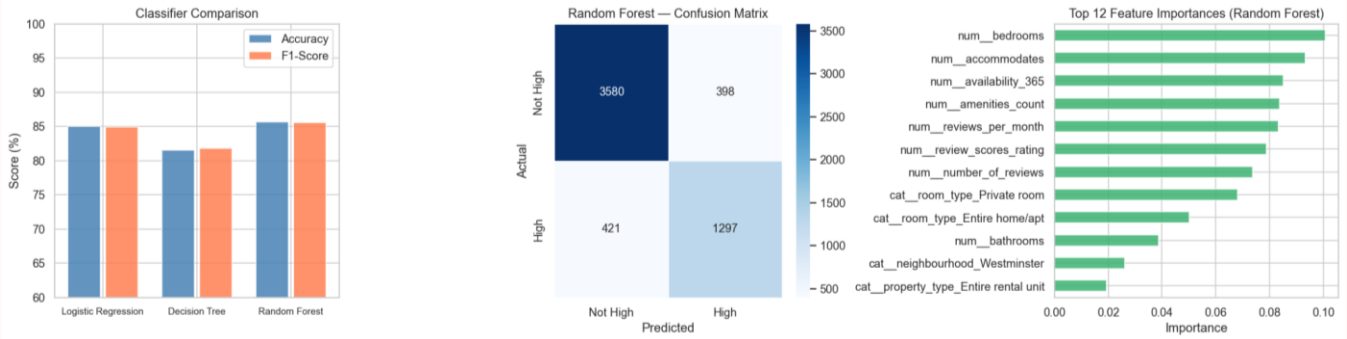


Figure 13: Classification results

Feature importance analysis from the Random Forest model identified the following top predictors:

- **Number of bedrooms** — strongest predictor; larger capacity correlates strongly with high-price classification
- **Accommodates** — closely related to bedrooms; reflects overall listing size
- **Neighbourhood encodings** — Central London boroughs drive high-price predictions
- **Room type** — entire home/apt flag is a strong positive predictor
- **amenities_count** — more amenities moderately predicts premium pricing

5.4 Regression — Predicting Nightly Price

Two regression models were evaluated for predicting continuous nightly price (after log-transformation of the target):

Model	RMSE (\$)	MAE (\$)	R ² Score	Notes
Linear Regression	\$39.45	\$27.74	0.6959	Baseline; assumes linearity
Gradient Boosting	\$37.75	\$26.46	0.7216	Best; captures non-linear price drivers

Table 5: Regression Model Comparison

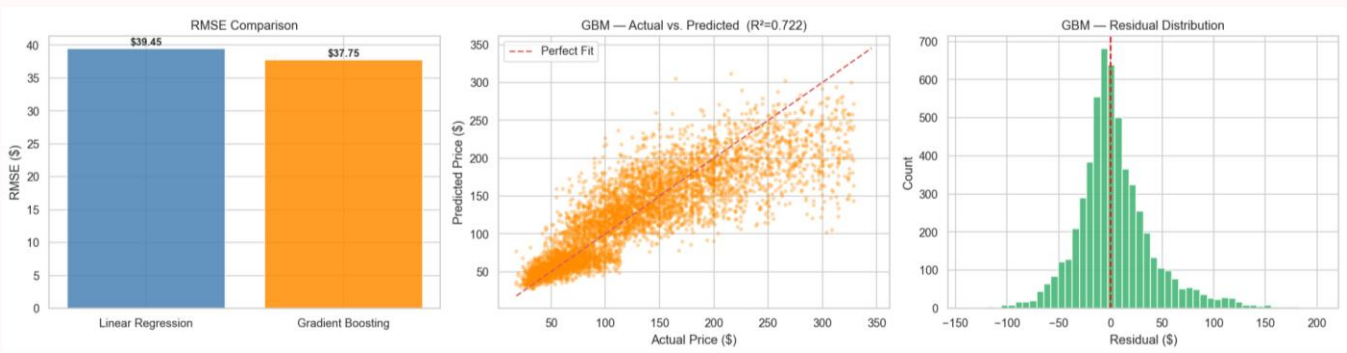


Figure 14: Regression results

5.5 Regression Model Interpretation

The Gradient Boosting Regressor achieved an R² of 0.7216, meaning it explains approximately 72% of the variance in nightly prices. The RMSE of \$37.75 represents the typical prediction error on original price scale, which is reasonable given the wide price range (\$10–\$437).

The residual distribution from Gradient Boosting is approximately centred at zero with a slight right tail, indicating the model occasionally under-predicts very high-price luxury listings — a common challenge when the upper tail of the price distribution is thin.

Linear Regression, constrained to linear relationships between features and log-price, underperforms significantly compared to Gradient Boosting. This confirms that the relationship between listing attributes and price is inherently non-linear.

6. Summary of Results

Table 7 consolidates the key outcomes from all four analytical techniques applied in this study.

Technique	Best Result	Key Finding
EDA	Median price: \$110/night	Entire home/apt dominates (65.6%); City of London is the most expensive borough (~\$201/night median); prices are right-skewed across all room types.
Association Rule Mining	79 rules, max Lift $\approx$ 2.87	95% confidence that budget price tier co-occurs with private rooms and small units; luxury tier is strongly driven by large units with high availability.
K-Means Clustering	K = 3, Silhouette = 0.4329	Three distinct market tiers: Standard Mid-Range (most stable), Premium High-Priced (lower relative reviews), and Economy Budget units.
Classification (RF)	Accuracy 85.62%, F1 85.59%	Bedrooms count and accommodates capacity are the top predictors of high-price classification; Random Forest outperforms Logistic Regression and Decision Tree.
Regression (GBM)	RMSE \$37.75, $R^2 = 0.7216$	Gradient Boosting captures 72% of price variance; outperforms Linear Regression by leveraging non-linear feature interactions.

Table 6: Summary of All Analytical Results

7. Conclusion

This study successfully demonstrates the application of the full KDD pipeline to the London Airbnb market. Using a representative dataset of 28,480 listings, four distinct analytical techniques were applied to extract meaningful insights at different levels of abstraction.

7.1 Key Takeaways

- **Market Structure:** The London Airbnb market is dominated by entire home/apartment listings (65.6%), with prices strongly influenced by neighbourhood, capacity, and room type. The median price of \$110/night masks significant geographic and property-level variation.
- **Pricing Patterns:** Association Rule Mining reveals clear structural patterns — budget pricing is overwhelmingly associated with private rooms and small units, while luxury listings cluster around large entire-home properties with high year-round availability.
- **Market Segmentation:** K-Means clustering identifies three actionable tiers — Economy Budget (~\$63), Standard Mid-Range (~\$114), and Premium High-Priced (~\$180) — each with distinct host strategies and guest profiles.
- **Predictive Power:** Random Forest classification predicts high-price listings with 85.62% accuracy, while Gradient Boosting regression explains 72% of nightly price variance ($R^2 = 0.7216$) with an RMSE of \$37.75.

7.2 Practical Implications

- **For Hosts:** Bedroom count and accommodation capacity are the most controllable levers for price positioning. Listing in central boroughs and offering entire home/apt significantly increases revenue potential.
- **For Guests:** Review scores are consistently high across all price tiers, confirming that affordable listings can deliver comparable satisfaction to premium ones.
- **For Platform Strategy:** The identification of three distinct market tiers necessitates tailored marketing approaches: value-driven messaging for Economy Budget listings, reliability and consistency framing for the Standard Mid-Range segment, and an emphasis on luxury and exclusive experiences for the Premium High-Priced tier.

7.3 Limitations & Future Work

- The current analytical framework is constrained by the nature of the static dataset snapshot, which precludes the modeling of high-frequency temporal dependencies, such as seasonal price volatility, intra-week demand fluctuations, or long-term longitudinal trends inherent in real-time market dynamics.
- Future work could incorporate geospatial features (proximity to tube stations, tourist attractions) and natural language processing of listing descriptions to further improve predictive accuracy.
- Time-series analysis of availability and price changes over calendar months could reveal seasonal demand patterns not captured in the current static snapshot.
- Further work should focus on deploying the predictive models into a real-time system for continuous monitoring and dynamic pricing recommendations, moving beyond the current static snapshot and batch processing approach.

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The End.